

NEURAL NETWORKS & DEEP LEARNING					
Semester	VII	Course Code		CEAM-04	Credits
Scheme of Instruction Hours/ Week		L	T	P	TOTAL
		3	0	2	42Hrs/Sem
Scheme of Examination TOTAL = 150 marks		TH	IA	TW	P O
		100	25	0	0 25

Prerequisites:

- Students should possess knowledge of basic linear algebra
- Knowledge of programming languages is a must.

Course Objectives:

The subject aims to introduce and equip students with knowledge on:

1	Understanding of concepts of neural network, design of logic functions and implementation of various learning rules using neural network.
2	Knowledge about multilayer neural network, unsupervised learning algorithms and associative memories.
3	An understanding of Deep Neural Networks such as Auto encoders, recurrent neural networks and convolutional neural networks.
4	Knowledge about Deep Neural Networks such as Recurrent and Recursive neural networks.

Course Outcomes:

At the end of course, student will be able to:

CO1	Describe concepts of neural network, design neural network to implement logic functions and solve problems related to various learning rules.
CO2	Explain working of multilayer neural network, its design considerations, implement clustering algorithms and associative memory networks for various applications.
CO3	Implement Deep Neural Network algorithms such as Auto encoders and Convolutional neural networks.
CO4	Design and implement Deep Neural Networks such as Recurrent and Recursive neural networks.

UNIT 1	
Introduction: Introduction to neural networks, structure of biological neuron, Mc-Culloch Pitts neuron model, Logic network realization by using Mc-Culloch Pitts neuron model, Neuron modelling for artificial neuron systems, Neural learning. Single layer network: Concept of linear seperability and non-linear separability. Training rules- Hebbian learning rule, perceptron learning rule, Delta learning rule, Widrow-Hoff learning rule and related problems. Multilayer network : Error back propagation algorithm or generalized delta rule.	12Hrs
UNIT 2	
Multilayer network Setting of parameter values and design considerations (Initialization of weights, Frequency of weight updates, Choice of learning rate, Momentum, Generalizability, Network size, Sample size, Non-numeric inputs), performance evaluation. Unsupervised learning: Clustering, simple competitive learning algorithm, LVQ algorithm, , SOM, Adaptive Resonance Theory. Associative memories: Hopfield networks, Brain-state-in-a-box network, and problems.	10 Hrs
UNIT 3	
Deep Neural Networks: Introduction & Necessity of deep neural networks (DNN), Auto encoder and its types. Convolutional Networks: The Convolution Operation - Variants of the Basic Convolution Function, Structured Outputs, Data Types, Efficient Convolution Algorithms, Random or Unsupervised Features - LeNet, AlexNet.	10 Hrs
UNIT 4	
Recurrent Neural Networks: Recurrent Neural Networks: Bidirectional RNNs - Deep Recurrent Networks Recursive Neural Networks: The Long Short-Term Memory and Other Gated RNNs. Deep Generative Models: Deep Belief networks, Boltzmann Machines, Deep Boltzmann Machine, Generative Adversial Networks.	10 Hrs

TEXTBOOKS	
1	Kishan Mehrotra, Chilukuri Mohan, Sanjay Ranka, Elements of artificial neural network, Edition-Second, Penram Publications.
2	J. Zurada; Introduction to Artificial neural network, Edition-Second, Jaico Publications
3	Ian Goodfellow, Yoshua Bengio, Aaron Courville; Deep Learning, Edition-Second, MIT Press
REFERENCES	
1	Satish Kumar, Neural networks: a classroom approach, Edition-Second, Tata McGraw-Hill
2	B. Yegnanarayana, Artificial Neural Networks, Edition-Second, Prentice Hall
3	S. Haykin, Neural Network and Learning Machines, Edition-Second, Prentice Hall

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(Minimum 08 Experiments to be performed from the following list)

Sr. No	Experiment
1	Study of different libraries for ANN and DNN
2	Design of logic gates using neural network
3	Implementation of Perceptron learning rule
4	Implementation of Delta learning rule
5	Implementation of classifier using EBPA
6	Implementation of clustering using SCL
7	Design of associative memory using Hopfield network
8	Implementation of Autoencoders-1
9	Implementation of Autoencoders-2
10	Implementation of Convolutional Neural Networks
11	Implementation of Recurrent Neural Networks
12	Implementation of Recursive Neural Network

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